

The way that we currently produce and consume generative AI *is* bad for the environment

Posted by u/nub0987654 • • 0 points • 16 comments

AI isn't inherently bad for the environment. We can certainly produce and consume it ethically. The reality, though, is that the vast majority of people do it in a way that does harm our environment, and it does need to be talked about.

According to Adam Zewe of MIT News, the sheer amount of electricity demanded to train most generative AI models and deploy them can increase carbon dioxide emissions and put pressure on the electricity grids.[1] This is because smarter LLMs often have tens of billions of parameters in order to tailor each response to each prompt.[2]

This electricity can only reasonably be generated through datacenters: centralized networks of computer servers to store, process, and distribute large amounts of, well, data.[3] Many of these are powered by fossil fuels or nonrenewable resources like natural gas,[2] and lots of fresh water is required to cool these servers.[4] Datacenters are not new by any stretch of the imagination, but the rise of AI rose their rate of construction dramatically.[1]

Right now, datacenters account for about 2% of global electricity consumption.[5] By 2040, the information and communication technology industry (which AI is a part of) is expected to comprise 14% of global carbon emissions, with the majority of those emissions stemming directly from artificial intelligence.[6] The typical ChatGPT query, for example, consumes about 10x as much energy as a typical Google search.[4]

That said, AI emissions are hard to generalize because much of it depends on the prompter's proximity to the datacenters, the kind of hardware used to run the AI, and the model being run. Each individual user's output is different,[2] and many users have negligible outputs.[7] However, 2.5 billion queries are sent to ChatGPT every single day, adding to 912.5 billion requests annually.[8] The sheer scale at which we are consuming AI makes it nearly impossible to manage properly.

AI is not uniquely bad for the environment, no. McDonald's is way worse of an offender than OpenAI could ever be. It isn't even entirely bad. However, I am of the mindset that where we can and where we are willing, we should cut down on stuff that, as a whole, harms more than it helps, like this very specific use case of artificial intelligence. We need to be responsible with our planet. And if that means one fewer Big Mac and one fewer image created with ChatGPT, then I say so be it. I love you all, and go drink some water :)

Works Cited

[1]: [Explained: Generative AI's environmental impact](<https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117>)

[2]: [Study reveals how much energy AI uses to answer your questions](<https://www.cnn.com/2025/06/22/climate/ai-prompt-carbon-emissions-environment-wellness>)

[3]: [The Climate and Sustainability Implications of Generative AI](<https://mit-genai.pubpub.org/pub/8ulgrckc/release/2>)

[4]: [AI's Impact on the Environment, Explained](<https://www.snopes.com/news/2025/01/16/ai-environment-carbon-footprint/>)

[5]: [Will AI help or harm the climate?](https://www.economist.com/podcasts/2025/04/09/will-ai-help-or-harm-the-climate)

[6]: [Artificial intelligence and climate change: ethical issues]
(https://www.emerald.com/jices/article/21/1/1/432616/Artificial-intelligence-and-climate-change-ethical)

[7]: [ChatGPT isn't great for the planet. Here's how to use AI responsibly.](https://www.washingtonpost.com/climate-solutions/2025/06/19/ai-chatgpt-energy-climate-change/)

[8]: [OpenAI says ChatGPT users send over 2.5 billion prompts every day]
(https://www.theverge.com/news/710867/openai-chatgpt-daily-prompts-2-billion)

Comments

u/YentaMagenta • 2 points • 2025-09-08 04:40 UTC

The big questions are: Will AI use really grow so much that it will become the majority of data center use? and Will the AI be useful?

The projections of how much energy AI will use are guesstimates at best, and much of this is based on marketing hype from AI companies. *Of course* AI companies want their investors to believe that AI use is going to be 10x, 20x more in the coming decades. But this is purely speculative, and so far we honestly haven't seen what the "killer app" is for AI that's really going to drive this.

This leads to the second question. Is what the AI doing useful? If AI is using 14% of global energy but is doing science and medicine that saves lives, reduces fossil fuel use, etc, then it's arguably totally fine for AI to be using that much energy. It's hard to imagine that AI will be using 14% of global energy without doing something that people find valuable. Maybe AI will get to the point where it's making custom feature length films for each individual, and that will be using a ton of energy for a frivolity, but we are *very* far from that.

What matters right now is how much does the AI use we are engaging in compare to other things we are doing. And AI is minuscule in its environmental impacts compared to everything else we do. If you want to tell people not to do something based on projections, then air travel should be the thing you focus on. If everyone in the world did as much air travel as people in the US, we'd cook the planet in no time. But no one says "Wow, we really need to stop this air travel thing" because of status quo bias and because they haven't decided that air travel is the new sexy tech we need to resist.

u/nub0987654 • 1 points • 2025-09-08 06:34 UTC

I'm actually largely swayed by your reasoning, apart from the idea that if anything, we should be going for air travel. Yes, we should definitely cut back on just how much we travel in the air, but air travel is fundamentally useful. There's little room for frivolity. Much of our use of AI, though, is purely for entertainment. AI, yes, *can* be used for science and medicine and good, beneficial things, but the vast majority of use *right now* is not in that ballpark. And generative AI does have an impact on the world, even if you see it as minuscule. That's why I'll largely lean anti-AI in the aspect of "we shouldn't use it frivolously", but I'll lean more neutral or even slightly pro on the fronts of "AI can be art and someone can be its artist" and "AI can and should be used in the fields of medicine and scientific innovation".

u/crossorbital • 8 points • 2025-09-08 04:04 UTC

> And if that means one fewer Big Mac and one fewer image created with ChatGPT, then I say so be it. I love you all, and go drink some water :)

Orrrrr you could skip two Big Macs instead, and do almost twice as much good?

I mean, Big Macs are shit anyway.

Seriously, if you actually want to reduce your personal environmental impact, you should pay attention to what actually makes a difference, not what's trendiest on social media. "AI is bad for the environment" is something you only say if you don't care that much about the environment but love engaging in performative nonsense.

u/nub0987654 • -1 points • 2025-09-08 04:09 UTC

You could skip two Big Macs and zero ChatGPT images. I'm not saying AI is uniquely bad, but it *is* bad for the environment, especially with the way we use it. Where we can and where we are willing, we should cut back on it. That's all I'm saying.

Also, one could argue that implying that our use of AI "doesn't make a difference" is engaging in performative nonsense itself. It is obviously impactful, whether that impact be good or bad.

u/EngineerBig1851 • 2 points • 2025-09-08 07:15 UTC

> you can skip two bigmacs and zero Chatgpt images

You can skip two thermonuclear bombs, and zero firecrackers.

u/nub0987654 • 1 points • 2025-09-08 07:16 UTC

I mean...true.

u/IndigoFenix • 2 points • 2025-09-08 06:51 UTC

Define "the way we use it."

You need to generate several **thousand** images before your energy consumption is even in the same ballpark as a human creating **one** image through traditional means. There ARE people doing that - generally automated ad systems - but people who take any kind of pride in their generative projects aren't.

Which means that the people getting targeted in art spaces for using AI are **not** the people whose energy consumption is a problem.

u/nub0987654 • 1 points • 2025-09-08 07:07 UTC

Uh...what?

> You need to generate several thousand images before your energy consumption is even in the same ballpark as a human creating one image through traditional means.

What do you mean by "creating one image through traditional means"? Are we talking pencil-and-paper, a drawing tablet, paint on canvas, or sticks and trees? What's an "image" here? And how could a single traditional image possibly use so much energy that it would take thousands of AI-generated images to match it? I ask this genuinely because I am quite confused by this assertion and would love clarification.

In any case, I agree that casual AI artists are not the problem and they should not be treated as such.

u/IndigoFenix • 1 points • 2025-09-08 07:35 UTC

The main energy consumer in the case of producing digital art traditionally is the **human**. As in, the energy cost of lighting and eating during the time you spend making the image. An image generator can produce images using much less energy than a human could, meaning that if viewed as a question of "how much energy did it cost to produce this image", the AI-generated image will be thousands of times less.

Of course, you're already using that energy to live even if you're not drawing, so using AI does consume additional energy. Just not very much of it.

The actual energy problem presented by AI could be thought of in terms of a "virtual population": How many extra people are you functionally representing during generation? For the average user, way less than one. Which makes sense, since those energy costs are represented by the amount you pay.

What AI does is essentially allow the conversion of electricity (and therefore money) into "virtual people" whose input-to-output ratio is much smaller than that of real people, but whose footprint can still add up if you're making millions of them.

The best solution to the problem is probably a carbon tax.

u/crossorbital • 3 points • 2025-09-08 05:55 UTC

Simply existing in western civilization is bad for the environment. Using AI isn't really appreciably worse than lots of other stuff people do. Is this bad? Yeah. Is avoiding AI specifically a useful way to help? Not really.

The only reason to focus on ChatGPT in particular is if someone dislikes it for unrelated reasons, and is trying to invent justifications after the fact.

So, no. The very fact that you're posting this here, as a way to push a "don't use ChatGPT" angle, shows that your argument is fundamentally unserious.

u/nub0987654 • 0 points • 2025-09-08 06:23 UTC

Well I'm not focusing on ChatGPT in particular. That was an example to illustrate my point about generative AI in general. Even so, I have stated that I'm not peddling any narrative that AI is particularly bad for the environment. I'm not trying to say generative AI is inherently bad, but the way we consciously choose to use it, even in some small part, harming the planet. All I'm saying is that if you are willing to do it, even if it, on such a small scale, does nothing, you should opt out of using generative AI. I've even cited sources. I don't know what more you could want from me

u/crossorbital • 1 points • 2025-09-08 06:32 UTC

What I want from you is for you to *actually* care about the environment, not about farming social media cred by chasing the most trendy cause to get those sweet imaginary internet points.

I don't expect to get that, mind you, but that's what I want.

u/nub0987654 • 0 points • 2025-09-08 06:43 UTC

That simply is not true. I am not chasing shiny internet points; in fact, with this post and many posts before it, I've been downvoted into oblivion for my (admittedly non-perfect) opinions and beliefs. Whether you want to attribute that to my lack of reasoning or not, it shows that I'm standing in a place that people here do not agree with.

I do care about the environment. You can not believe that, but I'm telling you the truth: I care about our planet and how we interact with it. I want to live in a world where the green, lush forests, the vast, sandy dunes, the frigid Arctic tundras, and humans coexist. I want people to cut down on their environmental emissions. I don't care if it's through culling their use of AI, or not eating the Big Mac they're craving, or boycotting meat companies, or advocating for the construction of pedestrian-friendly communities, or mailing their local officials about the excessive emission of carbon in their city, or whatever it may be.

Obviously, humans still gravitate to the path of least resistance, but I do care *actually* about the environment. I don't see how that label could be applied to me

u/crossorbital • 1 points • 2025-09-08 18:01 UTC

So it's entirely a coincidence that the very specific form of environmental harm reduction you focused on was not even close to the most damaging, but was instead the one most trendy to push on social media?

This is why I do not believe you. I'm willing to believe you're lying to yourself as well, but I simply do not

believe your real motivations are what you want them to be.

u/nub0987654

• 1 points
• 2025-09-08 20:09 UTC

It isn't a coincidence. This is literally *ai*wars. I'm not going to talk about the environmental harm of oil refineries on a subreddit about AI.

u/redditscraperbot2 • 4 points • 2025-09-08 05:42 UTC

I can count the number of big Mac's I've eaten in my 30+ years of life on one hand. I'm basically a saint.

u/nub0987654 • 1 points • 2025-09-08 03:32 UTC

Wow, I didn't know that Reddit didn't like links.

Let me see if I can fix this.

u/[deleted] • 1 points • 2025-09-08 03:29 UTC

[deleted]